Appendix

Compliance within regulatory gray areas: The case of the organic seed standard

Survey sample

1. Data collection method

- 2010: Data were collected by convenience sample only, shared by 23 organizations, including non-profits and foundations, organic certifiers, and departments of agriculture. Surveys were distributed primarily over email, but paper, phone, and in-person surveys were conducted at the request of the grower. We received a total of 1,027 responses (712 web-based, 315 paper-based), 546 of which included complete enough data for our analysis.
- 2015: Data were collected by both random and convenience sampling. For the random sample, 800 certified organic producers were randomly selected from the approximately 12,000 organic producers that use seed, recorded in the NOP database. To encourage responsiveness, three follow-up emails were sent following an initial email, paper copies were mailed on request, and individual phone calls were made to follow-up. From these 800, 197 (25%) responded (114 web-based, 83 paper-based) and 129 (16%) were complete enough to use in our analyses. For the convenience sample, surveys were shared by 41 organizations, primarily over email. We received a total of 1,148 responses from the convenience sample, 662 of which included complete enough data for our analysis. In total we had 791 usable responses.
- 2020: Data were collected by both random and convenience sampling using methods similar to 2015. For the random sample, 2,000 certified organic producers were randomly selected from the the NOP database and given the option to complete the survey over email or paper. After an initial email and postcard invitation, growers received follow-up contacts by email, postcard, and phone, and hard copies of the survey were sent. Survey distribution was managed by Washington State University's Social and Economic Sciences Research Center (SESRC) and distributed alongside the Organic Farming Research Foundation's (OFRF) National Organic Research Agenda. From these 2,000 surveys, 448 (22%) responded (206 web-based and 242 paper-based) and 191 (10%) were complete enough to use in our analyses. For the convenience sample, surveys were distributed by SESRC to all remaining certified organic operations over email, with postcard reminders. The survey was also advertised by Organic Seed Alliance (OSA) and OFRF. We received a total of 611 responses from the convenience sample, 217 of which included complete enough data for our analysis. In total we had 408 usable responses.

The breakdown between random and convenience sampling used in our analysis is summarized in Table S1. We consider the effect of sampling method, as well as survey distributor, through models in Section 2.

Table S1: Convenience and random samples, usable responses

Year	Total complete responses	Convenience responses (%)	Random responses (%)
2010	546	546 (100 %)	0 (0 %)
2015	791	662 (84 %)	129 (16 %)
2020	408	217 (53 %)	191 (47 %)

2. Checking for biases induced by data collection methods

In order to test how our data collection methods may bias the sample, we consider the effect of the sampling approach and survey distributor in our model. First we model the effect of sampling method with our 2015 and 2020 survey data (the only survey periods where we had both sampling approaches) by including a term for responses that were gathered by random sampling compared to convenience sampling (Table 2, Model 1). We see a small and insignificant effect for sampling method, and negligible changes in the results between the models run with and without the sampling consideration (Table 2, Model 2), suggesting no significant effect on results based on sampling approach.

Regarding concerns of survey distributor (i.e. that the distribution by OSA elicited responses from a select type of organic farmer), we model this effect with our 2010 and 2015 survey data (the only survey periods that identified specific distributors) by including a term for responses that were gathered through OSA channels compared to all others (Table 2, Model 3). We see a small and insignificant effect for distributor, and negligible changes in the results between the models run with and without the distributor consideration (Table 2, Model 4), suggesting no significant effect on results based on survey distributor.

Table S2: Beta regression model results including sampling and distributor effects

	Proportion of organic land planted with organic seed			
	Including sampling method (2015, 2020)	Without sampling method (2015, 2020)	Including survey distributor (2010, 2015)	Without survey distributor (2010, 2015)
	(1)	(2)	(3)	(4)
Barrier: Availability	−0.211***	−0.211***	−0.161***	−0.162***
	(0.042)	(0.042)	(0.042)	(0.042)
Barrier: Quality	−0.120***	−0.120***	−0.117***	−0.119***
	(0.045)	(0.045)	(0.041)	(0.041)
Barrier: Quantity	−0.054	−0.054	−0.008	−0.007
	(0.043)	(0.043)	(0.038)	(0.038)
Barrier: Form	0.035	0.035	0.013	0.014
	(0.040)	(0.040)	(0.037)	(0.037)
Year: 2020 (vs. 2015)	0.106	0.112		
	(0.084)	(0.081)		
Year: 2010 (vs. 2015)			−0.231***	−0.226***
			(0.077)	(0.077)
Values organic seed	0.189***	0.188***	0.199***	0.197***
	(0.041)	(0.041)	(0.036)	(0.036)
Farm size (log acres)	−0.268***	−0.268***	−0.267***	−0.263***
	(0.055)	(0.055)	(0.053)	(0.053)
Crop diversity	0.046	0.044	0.074*	0.075**
	(0.042)	(0.041)	(0.038)	(0.038)
Certifiers request organic seed use	−0.043	−0.043	−0.131*	−0.130*
	(0.077)	(0.077)	(0.073)	(0.073)
Seed saved or traded (%)	0.053	0.053	0.093***	0.091***
	(0.038)	(0.038)	(0.034)	(0.034)
Forage crops (vs. Field)	−0.971***	−0.971***	−1.011***	−1.013***
	(0.115)	(0.115)	(0.105)	(0.105)
Vegetable crops (vs. Field)	−0.609***	−0.611***	−0.589***	−0.594***
	(0.124)	(0.124)	(0.118)	(0.118)
Western region (vs. North Central)	−0.255**	−0.254**	−0.332***	−0.366***
	(0.103)	(0.103)	(0.102)	(0.098)
Northeastern region (vs. North Central)	−0.309***	−0.306***	−0.207*	−0.249**
	(0.106)	(0.105)	(0.118)	(0.113)
Southern region (vs. North Central)	−0.044	−0.045	−0.159	−0.181
	(0.136)	(0.136)	(0.131)	(0.130)
Random data collection (vs. convenience)	0.023			
	(0.090)			
Distributor: OSA (vs. Other)			−0.098	
			(0.081)	
Constant	1.058***	1.061***	1.165***	1.144***
	(0.090)	(0.089)	(0.088)	(0.086)
Observations	1,199	1,199	1,337	1,337
R^2	0.255	0.255	0.253	0.252
Log Likelihood	1,728.988	1,728.957	1,653.739	1,652.997

Note: *p<0.1; **p<0.05; ***p<0.01

3. Sample representativeness

While the mix of sampling methods creates a challenge for determining the sample's representativeness of the population, we aim to demonstrate that our sample is representative by comparing the geographic distribution of operations (Table S3) and farm size distribution (Figure S1) between our sample and USDA Census data.

We first use USDA Census data from 2011, 2016, and 2021 to align with our 2010, 2015, and 2020 survey periods. Across the three survey periods, the number of respondents represented approximately 6.3%, 5.9%, and 2.5% of the total number of organic farmers identified in the USDA Census in the related years. In Table S3 we compare the number of operations in our sample (n) to the population (N) by year and region. For each census period we calculated the standard deviation of the percentage of operations represented, which we used to identify regions that are over or under-represented. Over-represented year-region combinations are signaled by green text, while under-represented year-region combinations are signaled by red text. Only in 2010 do we see significant variation in representation across regions, with the North Central and Southern regions over-represented and the Northeast under-represented.

Table S3: Sample representativeness compared to USDA Censuses (2011, 2016, 2021)

Year	Region	Operations in sample (n)	Operations in USDA census (N)	Percent of operations represented (n/N)
2010	North Central	305	2986	10.2
2010	Northeast	51	1897	2.7
2010	South	51	540	9.4
2010	West	139	3296	4.2
2015	North Central	386	4689	8.2
2015	Northeast	141	3246	4.3
2015	South	75	1096	6.8
2015	West	189	4529	4.2
2020	North Central	174	5824	3
2020	Northeast	98	3888	2.5
2020	South	38	1209	3.1
2020	West	98	5273	1.9

Next, to understand farm size representativeness, we look at USDA Census organic tabulations, reported only for the 2012 and 2017 censuses.

In Figure S1 we compare these USDA Census data against our survey sample. These comparisons show a consistent over-representation of the smallest farms (1-9 acres), especially in the North Central and South, and under-representation of the largest farms. Still, our sample captures a distribution of organic farm sizes that includes medium and large farms, allowing us to understand the effect of farm size in our analysis. Additionally, an important consideration is that we calculate farm size as reported acreage planted to three major crop categories, but this approach likely underestimates acreage generally compared to USDA Census data, which reports all certified land. Furthermore these data are not crop specific and so USDA data also includes non-seed crop operations (i.e. orchards).

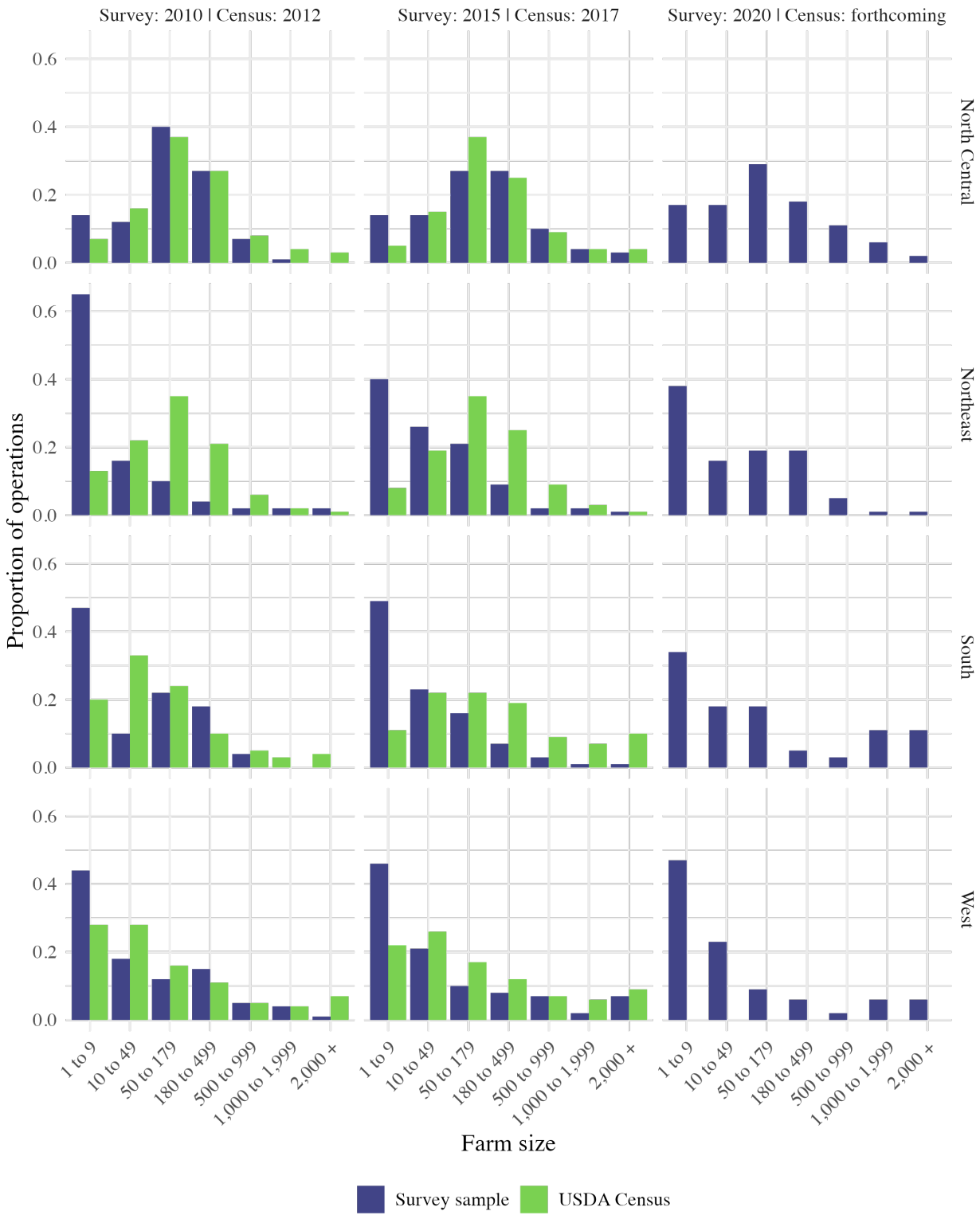


Figure S1: Sample farm size distribution compared to USDA Census data

Model checks

4. Full model results (Table S4)

Table S4: Beta regression model results

	Proportion of production acreage planted with organic seed
Barrier: Availability	-0.193^{***}
	(0.036)
Barrier: Quality	-0.121^{***}
	(0.036)
Barrier: Quantity	-0.022
	(0.034)
Barrier: Form	0.036
	(0.032)
Year: 2010 (vs. 2015)	-0.213^{***}
	(0.076)
Year: 2020 (vs. 2015)	0.136^{*}
	(0.080)
Values organic seed	0.180^{***}
	(0.032)
Farm size (log acres)	-0.112^{***}
	(0.021)
Crop diversity	0.078^{**}
	(0.034)
Certifiers request organic seed use	-0.092
	(0.064)
Seed saved or traded (%)	0.098^{***}
	(0.031)
Forage crops (vs. Field)	-0.890^{***}
	(0.093)
Vegetable crops (vs. Field)	-0.515^{***}
	(0.103)
Western region (vs. North Central)	-0.368^{***}
	(0.086)
Northeastern region (vs. North Central)	-0.287^{***}
	(0.093)
Southern region (vs. North Central)	-0.150
	(0.113)
Constant	1.063^{***}
	(0.078)
Observations	1,745
R^2	0.237
Log Likelihood	2,259.708
Note:	*p<0.1; **p<0.05; ***p<0.01

5. Considering non-allowable barriers (Table S5)

Table S5: Beta regression model results including non-allowable barriers

	Proportion of production acreage planted with organic seed
Barrier: Availability	-0.166^{***}
	(0.038)
Barrier: Quality	-0.113^{***}
	(0.037)
Barrier: Quantity	-0.005
	(0.035)
Barrier: Form	0.064^{*}
	(0.033)
Barrier: Price	-0.075^{**}
	(0.034)
Barrier: Buyer requirement	-0.118^{***}
	(0.034)
Year: 2010 (vs. 2015)	-0.195^{**}
	(0.077)
Year: 2020 (vs. 2015)	0.157^{*}
	(0.081)
Values organic seed	0.161^{***}
	(0.033)
Farm size (log acres)	-0.103^{***}
	(0.021)
Crop diversity	0.079^{**}
	(0.034)
Certifiers request organic seed use	-0.057
	(0.065)
Seed saved or traded (%)	0.102^{***}
	(0.032)
Forage crops (vs. Field)	-0.890^{***}
	(0.094)
Vegetable crops (vs. Field)	-0.494^{***}
	(0.104)
Western region (vs. North Central)	-0.380^{***}
	(0.086)
Northeastern region (vs. North Central)	-0.302^{***}
	(0.094)
Southern region (vs. North Central)	-0.166
	(0.115)
Constant	1.027^{***}
	(0.079)
Observations	1,716
R^2	0.246
Log Likelihood	2,240.125
Note:	*p<0.1; **p<0.05; ***p<0.01

6. Mediation analyses

To check for mediating effects between values, barriers, and operational attributes, we run three mediation analyses to determine if one variable (a mediator) is transmitting the effect of another variable.

We first test the potential mediation effect of larger farms on valuing organic seed. In other words, are those who value organic seed more likely to have smaller farms, and might this influence model results? We depict a simplified representation of this mediating effect in Figure S2.

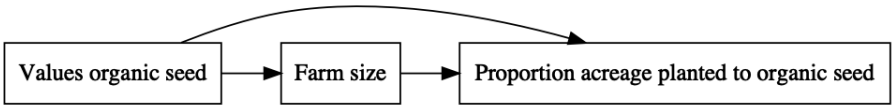


Figure S2: Mediation diagram: Organic seed value and farm size

We run a mediation analysis by comparing results of our full regression model (e.g. results in Table S4) with results of a model that regresses the same variables on the potential mediator: farm size. Results are presented in Table S6. The Total Effect of organic seed value without considering farm size is 0.052, while the average direct effect (ADE), which considers the effect of the mediatior, lowers this effect size to 0.046. This is reflected in the Average Causal Mediation Effect (ACME), is small but significant.

Table S6: Mediation analysis of farm size on quantity barrier

Effects	Estimate	p-value
ACME (avg)	0.006	< 0.001
ADE (avg)	0.046	< 0.001
Prop. Mediated (avg)	0.110	< 0.001
Totel Effect	0.052	< 0.001

Next we address the potential mediation effect of experiencing organic seed quantity barriers on larger farms (Figure S3). In other words, are larger farms be more likely to experience the barrier of limited organic seed quantity, and might this influence model results?

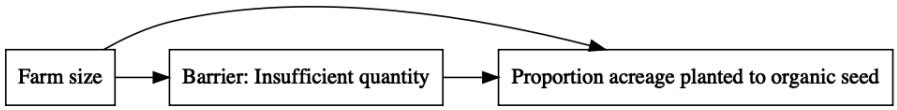


Figure S3: Mediation diagram: Farm size and organic seed quantity

Again, we run a mediation analysis by comparing results of our full regression model with results of a model that regresses the same variables on the potential mediator: quantity barrier. Results are presented in Table S7, where the ACME is not significant, signalling that the quantity barrier is not mediating the effect of farm size.

Table S7: Mediation analysis of farm size on quantity barrier

Effects	Estimate	p-value
ACME (avg)	-0.002	0.454
ADE (avg)	-0.086	< 0.001
Prop. Mediated (avg)	0.018	0.454
Totel Effect	-0.088	< 0.001

Last, we addresses the potential mediation effect of experiencing organic seed availability barriers on more diverse farms (Figure S4).

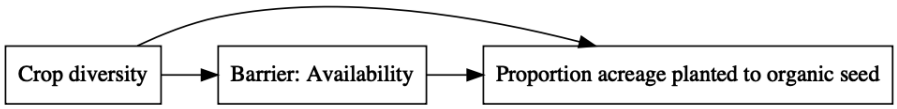


Figure S4: Mediation diagram: Crop diversity and organic seed availability

Much like the analysis concerning farm size and the quantity barrier, we find that the availability barrier is not influencing the effect of crop diversity (Table S8).

Table S8: Mediation analysis of crop diversity on availability barrier

Effects	Estimate	p-value
ACME (avg)	-0.002	0.142
ADE (avg)	0.028	0.014
Prop. Mediated (avg)	-0.090	0.156
Totel Effect	0.026	0.018

7. Predicted organic seed use (Figures S5 and S6)

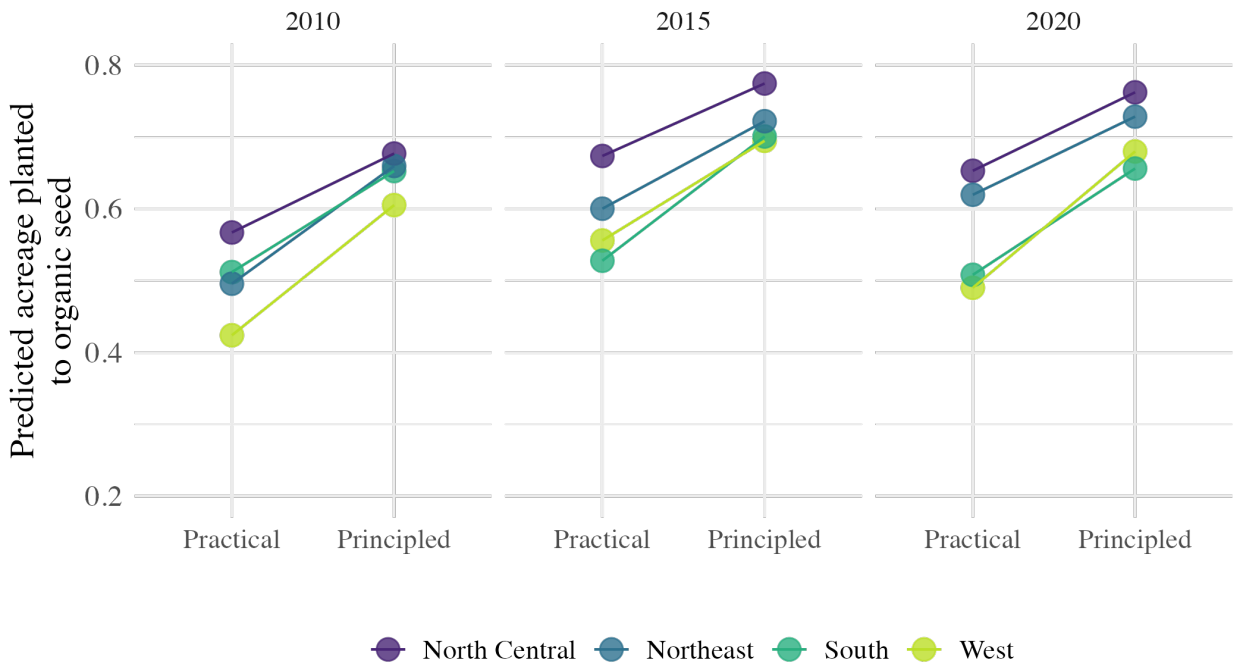


Figure S5: Predicted organic seed use for field crop growers of two farmer profiles.

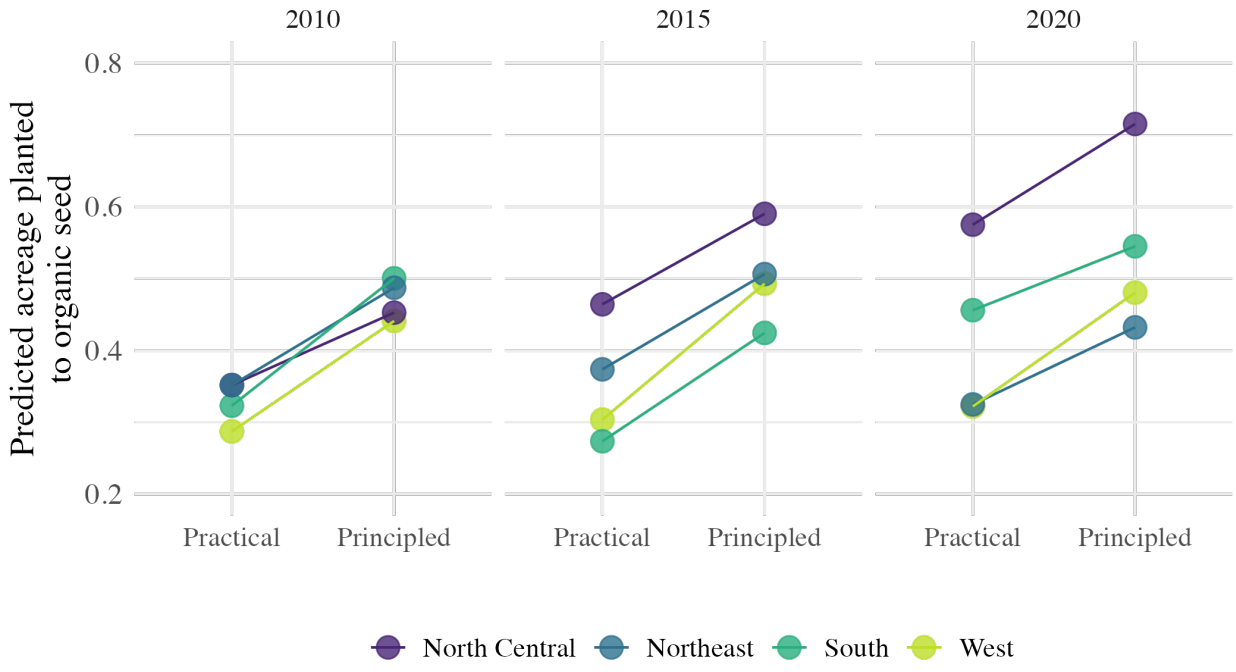


Figure S6: Predicted organic seed use for forage crop growers of two farmer profiles.

8. Self-reported behavior change

Results from a multinomial regression below (Table S9) are based on the question, *Over the last three years have you increased the percentage of certified organic seed that you use?* The baseline response is "No."

Table S9: Multinomial regression model results

	Self-reported increase in organic seed usage compared to baseline:	
	No: stayed about the same amount or decreased	
	Yes: Increased the %	Already at 100%
Barrier: Availability	0.101	−0.547***
	(0.073)	(0.090)
Barrier: Quality	0.094	−0.318***
	(0.066)	(0.111)
Barrier: Quantity	0.086	0.216**
	(0.065)	(0.099)
Barrier: Form	−0.040	−0.051
	(0.058)	(0.103)
Year: 2010 (vs. 2015)	1.185***	−4.537***
	(0.143)	(1.014)
Year: 2020 (vs. 2015)	−0.404***	−0.335*
	(0.150)	(0.190)
Values organic seed	0.316***	0.424***
	(0.060)	(0.101)
Farm size (log acres)	0.035	−0.217***
	(0.039)	(0.058)
Crop diversity	−0.011	−0.024
	(0.063)	(0.093)
Certifiers request organic seed use	0.654***	0.153
	(0.118)	(0.177)
Seed saved or traded (%)	−0.150**	0.050
	(0.062)	(0.079)
Forage crops (vs. Field)	0.231	−0.605**
	(0.175)	(0.245)
Vegetable crops (vs. Field)	0.815***	−1.341***
	(0.198)	(0.296)
Western region (vs. North Central)	−0.157	−0.186
	(0.158)	(0.239)
Northeastern region (vs. North Central)	−0.104	−0.370
	(0.173)	(0.249)
Southern region (vs. North Central)	−0.220	0.285
	(0.213)	(0.304)
Constant	−0.635***	−0.214
	(0.152)	(0.190)
Observations: Baseline (611)	841	265

Note: *p<0.1; **p<0.05; ***p<0.01